

TROY LORENTS

Senior Full-Stack & AI Engineer

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PROFESSIONAL SUMMARY

Full-stack and AI engineer with 8+ years of building production, cloud-native systems and the last 5+ years focused on AI-powered software, intelligent automation, and enterprise workflow optimization. Experienced delivering end-to-end solutions across React/TypeScript frontends, Python (FastAPI) and .NET backends, and scalable data platforms. Designed and deployed multi-agent systems, RAG pipelines, and production AI platforms using Azure AI Foundry, OpenAI, and Claude to automate business processes, reduce operational bottlenecks, and improve organizational efficiency.

TECHNICAL SKILLS

AI & ML: Azure OpenAI, Azure AI Foundry, Azure AI Search, Azure Document Intelligence, Anthropic Claude, Agentic & Multi-Agent Systems, RAG & Agentic RAG, Hybrid Search (BM25 + Vector), LangChain/LangGraph, Semantic Kernel, LLM Evaluation & Agent Evaluation Harnesses, PyTorch

Languages: Python, C#, TypeScript, JavaScript, SQL, Bash, PowerShell

Frontend: React, Next.js, Vue, React Native, TanStack, Material UI, Tailwind CSS

Backend: .NET Core, FastAPI, Flask, Node.js/Express, REST, GraphQL, Docker, Kubernetes, SDK Design & Development

Data: SQL Server, PostgreSQL, MySQL, Cosmos DB, DynamoDB, MongoDB, Neo4j, Redis, Qdrant

Cloud & DevOps: Azure (App Service, DevOps, API Management, AI Search, Cosmos), AWS (Lambda, EC2, ECS/Fargate, API Gateway, S3), CI/CD, GitHub Actions, Cloudflare, CircleCI, Octopus Deploy

Automation & Integration: n8n, Power Automate, UiPath, ServiceNow, Airtable, Stripe, REST APIs

PROFESSIONAL EXPERIENCE

Arizona State University | Web Application Developer (Full-Stack & AI) 2024 - Present

School of Computing & Augmented Intelligence, Ira A. Fulton Schools of Engineering

- Built SAMS, an end-to-end student hiring platform replacing a 150–200-hour manual process; serves 300+ users, cut processing time ~80%, and is being piloted by other ASU colleges. Led rollout backed by HR leadership and the Dean.
- Re-architected AWS hosting onto a single EC2 Windows Server instance running 200+ student .NET projects, each team container-isolated, cutting the monthly bill from \$850 to \$60 (93% reduction).
- Built production AI features using Python/FastAPI, frontier model APIs, Azure AI Foundry and LLM APIs: RAG with document ingestion, vector search, model routing across 17+ LLMs, tool-calling agentic workflows, and a conversational hiring chatbot.
- Designed and maintained SQL Server, PostgreSQL, and Cosmos DB data models supporting transactional and AI-driven workloads.
- Deployed across Azure (DevOps, App Service, API Management) and AWS (Lambda, API Gateway, S3, CloudWatch); automated cross-team handoffs with n8n, UiPath, Power Automate, Power Apps, and ServiceNow.
- Implemented modern authentication and authorization using OAuth 2.0, CAS SSO, and RBAC across enterprise applications and AI services, enabling secure API access and role-based permissions.

Zywave | Software Engineer 2018 - 2024

Healthcare Data SaaS, Scottsdale, Arizona

- Engineered enterprise healthcare SaaS with React, TypeScript, C#, ASP.NET Core, Entity Framework, and Dapper, supporting 10,000+ users with HIPAA-compliant architecture and secure API standards.
- Built an API quoting adapter for an insurance carrier in C#/ASP.NET Core (LINQ, Dapper, Swagger/OpenAPI) with full unit coverage in MOQ and Fluent Assertions.
- Designed databases across MySQL, SQL Server, MariaDB, and DynamoDB; tuned SQL and async processing to reduce latency and production incidents.
- Owned AWS infrastructure (EC2, S3, Lambda, DynamoDB, CloudWatch) and containerized CI/CD with Docker, Rancher, GitLab, and Octopus across QA, PRE, and DEV environments.

RateFactory (acquired by Zywave) | Data Administrator 2017 - 2018

- Built and ran ETL pipelines on large insurance datasets with Python (Pandas, OpenPyXL), Excel VBA, and uBot Studio web scrapers.
- Led a team of four administrators on task planning, onboarding, and QA across SQL Server, PostgreSQL, and DynamoDB.

PROJECTS

- **useKnockout API:** Built a production serverless GPU inference API using FastAPI, Modal, PyTorch, and BiRefNet, delivering automatic scale-to-zero deployments and published multi-language SDKs (Node.js, Python, React, CLI) to npm and PyPI.
- **Agent OS:** Open-source multi-agent orchestration platform built with FastAPI and React/TypeScript, supporting autonomous multi-agent workflows, secure REST APIs, PyTorch-based ML experimentation, Cloudflare, Neo4j, Redis, and Qdrant for graph, memory, and vector retrieval.
- **Resume Match AI:** Designed and built a full-stack AI platform using React, Flask, Azure OpenAI, Azure AI Search, Azure Cosmos DB, and Azure Document Intelligence, implementing production RAG pipelines, hybrid BM25 + vector search with RRF ranking, semantic skill extraction, embeddings, and REST APIs for intelligent resume-to-job matching
- **ATLAS:** Architected a multi-tenant AI SaaS platform using Python (FastAPI), React, PostgreSQL, Azure OpenAI, and specialized AI agents to automate business operations through role-based workflows, human-in-the-loop approvals, and cloud-native microservices.

Additional projects at troy-lorents-portfolio.netlify.app

EDUCATION & CERTIFICATIONS

B.S. Software Engineering, Ira A. Fulton Schools of Engineering, Arizona State University

B.A. Business Law, W. P. Carey School of Business, Arizona State University (May 2018)